

FEB12001X – Resit 7 July 2026

FEB12001X Applied Microeconomics 25/26 (ESE) · 10 exercises · 60.0 points

Question 1 - Public economics #63968234

9 pts · Last updated 13 July, 09:20

TEXT BLOCK

Suppose there are two individuals, Anne and Ben, who both derive utility from the consumption of a public good. Anne's valuation of the good in euro is given by

$$U_A = a q_A - \frac{1}{2} (q_A)^2,$$

where q_A is the quantity consumed by Anne. Ben's valuation in euro is given by

$$U_B = b q_B - \frac{1}{2} (q_B)^2,$$

where q_B is the quantity consumed by Ben. Supply of the public good on the market is perfectly elastic. Marginal production cost are equal to the price $p = MC = c$ euro. We assume that $a > b > c$ and that $b + c > a$.

2 PTS · OPEN · 1/2 PAGE

Derive the socially efficient level of consumption of Anne and Ben.

Model answer

$$MU_A + MU_B = MC \text{ gives } a + b - 2q = c \rightarrow q = \frac{1}{2} (a + b - c)$$

Use of $MU_A + MU_B = MC$

1 pt

correct calculation of marginal utilities

1 pt

TEXT BLOCK

Suppose that the public good is provided by the market. Anne and Ben decide independently how many units of the public good to buy (of course taking the presence of the other into account).

2 PTS · OPEN · 3/5 PAGE

Determine how many units of the public good Ben consumes in the Nash-equilibrium that arises on the market.

Model answer

In the Nash equilibrium Anne provides $a - c$ units and Ben 0, such that Ben consumes $a - c$ units. If Ben provides 0, it is optimal for Anne to provide such that

$MU_A = c \rightarrow a - q_A = c \rightarrow q_A = a - c$ units. Then $MU_B = b - (a - c) < c$, which implies that Ben optimally provides 0. This is the only Nash equilibrium.

Correct Nash equilibrium with correct conclusion regarding Ben's consumption

2 pts

Correct Nash equilibrium, but with conclusion that Ben consumes 0.

1 pt

Any answer where Ben provides (buys) a positive amount, including $q_B = b - c$.

0 pts

TEXT BLOCK

Now suppose the government offers a subsidy per unit of production to producers of the public good. The Pigouvian level of the subsidy equals $\frac{1}{2}(b + c - a)$ euro.

3 PTS · OPEN · 1/2 PAGE

Explain (in words) why the Pigouvian subsidy increases in b . Use no more than three sentences.

Given that Anne provides in the Nash equilibrium, the marginal utility of Ben is the marginal external benefit. 1 pt

The MEB increase in b (or: is higher if b is higher) 1 pt

to make Anne internalise Ben's benefits, the Pigouvian subsidy needs to increase in b as well. 1 pt

TEXT BLOCK

In case of pure public goods, at least one of the conditions of the Coase Theorem is always violated.

2 PTS · OPEN · 1/2 PAGE

Which of the characteristics of a pure public good makes that at least one of the conditions of the Coase Theorem is always violated? Explain why. Use no more than four sentences.

Model answer

- Best answer: A pure public good is non-excludable. This makes it impossible (= too costly) to enforce a contract where people promise to contribute to (pay for) the provision of the public good if they consume it. Hence, it violates the condition that transaction cost should be low.

- Alternative answer: A pure public good is non-excludable. This makes it impossible to enforce property rights, in the sense that an owner of a public good cannot prevent other people from using it (for free).

- Alternative answer: A pure public good is non-rival. This makes that all individuals that value the public good should consume it. The transaction cost of bargaining with many individuals is too large.

Note: it is possible that pure public goods have an owner (think of a lighthouse or a dike). Hence, the condition that property rights should be assigned is not necessarily violated in case of public goods.

Correct reasoning with proper use of relevant characteristics and conditions 2 pts

Any answer that just lists both characteristics of a public good and from that concludes that it violates at least one of the conditions of the CT, without proper explanation. 0 pts

Question 2 - Public economics #50629717

7 pts · Last updated 26 June, 08:36

TEXT BLOCK

Bert has 200 hours per month which he has to divide between working and leisure. His net wage is 20 euro per hour. Leisure L is the number of hours (out of the 200) that he does not spend working (hence, $L = 200 - \text{number_hours_work}$). The government has introduced the following system of poverty alleviation: People who have less than 500 euro per month earnings from work receive a grant of 600 euro per month. For people who earn more than 500 euro per month, the grant is reduced by 50 cents for every euro earnings from work above 500 euro. People who earn 1700 euro or more from work receive no grant.

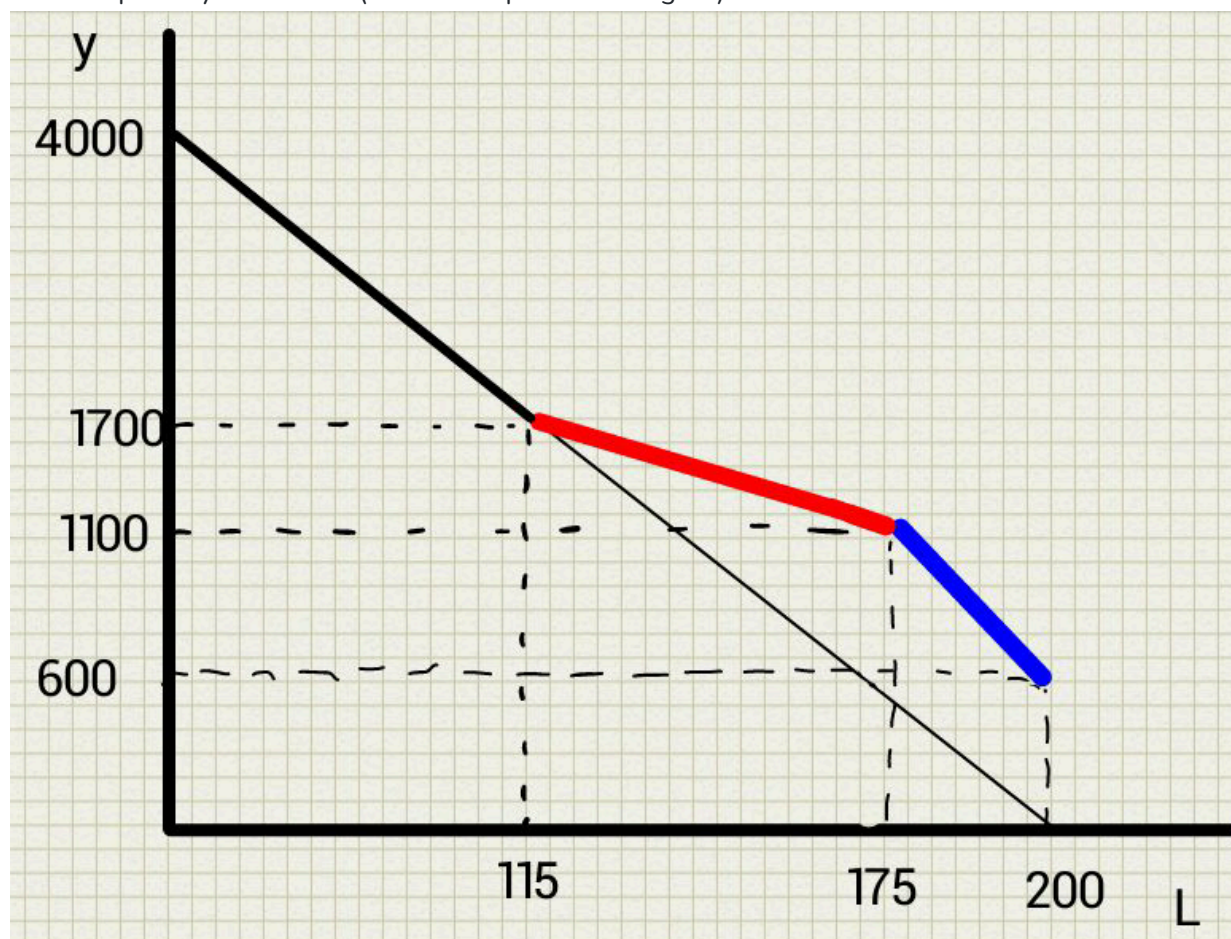
You can ignore taxes throughout this exercise.

3 PTS · OPEN · 3/5 PAGE

Draw Bert's budget constraint, with total net income on the Y-axis and number of hours of leisure on the X-axis. Clearly indicate the relevant points on the budget constraint on both axes (with numbers).

Model answer

See the figure below. Without system of poverty alleviation, Bert's budget constraint is $Y=4000-20L$. With poverty alleviation system, if Bert does not work at all ($L=200$) his income is 600 euro. His income raises with 20 euro for every hour he works until he earns 500 euro income for labour. This happens after 25 hours of work and hence total income is 1100 euro ($L=175$, blue part of BC). Next, every hour of work effectively yields only 10 euro additional income (because his grant is reduced by 10 euro if he works an additional hour), until the grant is zero. This happens when his total earnings from work are 1700 euro, at $1700/20 = 85$ hours or work ($L=115$; red part of BC). From there onwards, an hours of work yields 20 euro extra income (identical to BC without poverty alleviation system, thick black part of the BC). Note: the thin black line is not part of the BC with poverty alleviation (but can be part of the figure).



both endpoints Budget constraint correct: ($L=0, Y=400$) and ($L=200, Y=600$)

1 pt

Budget constraint with three parts

1 pt

At least one kink correct (either ($L=115, Y=1700$) or ($L=175, Y=1100$))

1 pt

TEXT BLOCK

Bert has diminishing marginal utility of both income and leisure (such that his indifference curves relating to the trade-off between income and leisure have the normal shape (convex)). Suppose that if the government would not have a system of poverty alleviation (grant is 0 for everyone), Bert would choose to work 85 hours ($L=115$).

2 PTS · OPEN · 1/2 PAGE

Argue whether Bert works more than 85 hours, less than 85 hours, or exactly 85 hours given that the government has introduced the poverty alleviation system as described above. Argumentation is crucial. (Hint: Calculations are neither necessary nor possible. You can use a new sketch of Bert's budget constraint in you argumentation.)

Model answer

If Bert works 85 hours when there is no poverty alleviation system, his income is 1700 euro. Graphically, this implies that an indifference curve in that point touches the BC without poverty alleviation system. (Note: we do not know how the IC runs exactly, only that it is convex and touches the BC in this point. This is enough to answer the question). This implies that in this point, an extra hour of leisure is worth as much to Bert as 20 euro. With the poverty alleviation system, the same point is possible ($L=115$, $Y=1700$). However, in that point, an hour of leisure has an (opportunity) cost of only 10 euro (because if he works an hour less, he receives a grant of 10 euro, red part of the BC). As he values an hour of leisure at 20 euro in that point, he optimally chooses to work less. Graphically: the indifference curve that touches BC without poverty alleviation system at $L=115$, $Y=1700$, lies beneath the BC with poverty alleviation system at higher levels of L (as the slope of the IC equals the slope of the BC without poverty alleviation system, and this BC is steeper than the BC with poverty alleviation system from this point (red part of BC in figure above)).

Indifference curve in ($L=115$, $Y=1700$) touches the BC without poverty alleviation system, or marginal valuation of leisure in that point equals 20 euro. With the poverty alleviation system, the same point is possible ($L=115$, $Y=1700$). 1 pt

With the poverty alleviation system, in that point, an hour of leisure has an (opportunity) cost of only 10 euro. Or: with the poverty alleviation system, working becomes less attractive for $L > 115$. 1 pt

2 PTS · OPEN · 3/5 PAGE

Suppose you would know Bert's utility function. Then, it would be possible to calculate the loss of efficiency due to Bert's response to the introduction of this poverty alleviation system. What steps must you take to calculate this? Note: you do not need to perform the calculation. It is not sufficient to mention terms; you must clearly indicate the steps required to carry out this calculation.

Determine the grant actually received after introduction of the poverty alleviation system 0.5 pts

Determine the utility derived by Bert after introduction of the poverty alleviation system 0.5 pts

Determine the equivalent variation: how much money would you have to give Bert in the old situation to achieve this level of utility. 0.5 pts

The loss in efficiency is the difference between the actual benefit received and the equivalent variation. 0.5 pts

Question 3 - Public economics #63964769

10 pts · Last updated 24 June, 11:30

TEXT BLOCK

Consider the market for good X . Demand for good X is given by

$$\backslash[$$

$$P=200-\frac{1}{2}Q_{\{D\}}$$

$$\backslash]$$

where P is the price of good X (in euro) and Q_D is quantity demanded. Supply of good X is given by

$$\backslash[$$

$$P=20+Q_{\{S\}}$$

$$\backslash]$$

where Q_S is quantity supplied. Consumption of good X leads to external cost. The marginal external cost are given by

$$\backslash[$$

$$MEC=\frac{1}{2}Q_{\{D\}}$$

$$\backslash]$$

Suppose there are many identical consumers of good X . Furthermore, suppose that the external cost of the consumption of good X are fully incurred by consumers of good X .

2 PTS · OPEN · 1/5 PAGE

Explain (in words) why market failure arises on the market for good X , even if the people that are hurt by the external cost are consumers of good X . Use no more than two sentences.

The external cost of the consumption of an individual consumer falls on other consumers 1 pt

These cost on other consumers are not taken into account by the individual consumer in the decision whether / how much to consume. 1 pt

2 PTS · OPEN · 2/5 PAGE

Determine the (economic) tax incidence of a tax of 30 euro per unit levied on the consumers of good X .

Model answer

Equilibrium without tax: $200 - \frac{1}{2}Q = 20 + Q$, such that $Q = 120$ and $P = 140$. With tax, we get $200 - 30 - \frac{1}{2}Q = 20 + Q$, such that $Q = 100$.

Consumers pay 150 and producers receive 120 euro. Hence, producers bear $2/3$ of the tax burden.

both pre and post-tax equilibrium correct 1 pt

Alternative: correct calculation of demand and supply elasticities.

Given equilibria, correct way of determining distribution of tax burden 1 pt

Alternative: correct use of demand and supply elasticities in determining tax incidence.

2 PTS · OPEN · 1/5 PAGE

Explain the effect of a higher elasticity of supply of good X on the tax incidence. Use no more than two sentences.

A higher elasticity of supply implies that suppliers have better alternatives. 1 pt

This allows them to shift a larger part of the tax burden to consumers. 1 pt

2 PTS · OPEN · 3/5 PAGE

Determine whether a unit tax of 30 euro increases or decreases efficiency on the market for good X , compared to the situation without a tax on good X .

Model answer

Various ways to solve this. One way: Without tax, $Q = 120$. Consumer surplus equals $\frac{1}{2} (200 - 140) * 120 = 3600$. Producer surplus equals $\frac{1}{2} (140 - 20) * 120 = 7200$. External cost are $\frac{1}{2} (60) * 120 = 3600$. Hence, total value created equals $3600 + 7200 - 3600 = 7200$ euro.

With tax: Consumer surplus equals $\frac{1}{2} (200 - 150) * 100 = 2500$. Producer surplus equals $\frac{1}{2} (120 - 20) * 100 = 5000$. External cost are $\frac{1}{2} (50) * 100 = 2500$. Tax revenue equals $30 * 1000 = 3000$. Hence, total value created equals $2500 + 5000 - 2500 + 3000 = 8000$ euro. Hence total value

created increases with 800 euro.

Alternative approach is to integrate demand - supply - external cost between $Q = 120$ and $Q = 150$

Beyond the question: note that the decline in producer surplus is exactly twice the decline in consumer surplus, which is another way of showing the tax incidence of question a.

correct 2 pts

2 PTS · OPEN · 3/10 PAGE

Determine the level of a Pigouvian tax on good X .

Model answer

Socially efficient level: $200 - \frac{1}{2} Q = 20 + Q + \frac{1}{2} Q$, such that $Q = 90$. Substitute in MEC gives Pigouvian tax of 45 euro. Check: $200 - 45 - \frac{1}{2} Q = 20 + Q$ gives $Q = 90$.

Correct calculation of socially efficient level 1 pt

Substituting the socially efficient level in MEC to determine the Pigouvian tax 1 pt

Question 4 - Public economics #63973326

4 pts · Last updated 24 June, 12:54

TEXT BLOCK

Suppose that the citizens of a country differ in productivity α . This productivity α is uniformly distributed between 0 and 1. A citizen with productivity α who has a job earns income $Y = 2500\alpha$ euro. A citizen without a job earns income $Y = 225$ euro. Utility of income is given by $U = \sqrt{Y}$. Searching for a job is difficult and time-consuming, and yields a loss in utility equal to c . If a citizen searches for a job, the probability that the citizen finds a job is given by π . A citizen who does not search never gets a job. Citizens maximise their expected utility. The government considers introducing the following system of poverty alleviation. Citizens who earn less than 625 euro receive a benefit that increases their total income to 625 euro (such that the benefit equals $625 - \text{own_earnings}$). The government cannot observe an individual's productivity α , but can observe earnings and whether citizens search for a job. Throughout the exercise, you can ignore taxes.

2 PTS · OPEN · 2/5 PAGE

Suppose $\pi = 0.8$ and $c = 8$. Determine which citizens search for a job before the introduction of the benefit system.

Model answer

A citizen searches for a job if $\pi \sqrt{2500\alpha} - c + (1 - \pi) \sqrt{225} \geq \sqrt{225} \rightarrow \pi (50 \sqrt{\alpha} - 15) \geq c$. Substituting for $\pi = 0.8$ and $c = 8$ yields $40 \sqrt{\alpha} - 12 \geq 8$, yielding $\sqrt{\alpha} \geq \frac{20}{40} \rightarrow \alpha \geq 0.25$.

Expected utility with search and utility without search correctly determined 1 pt

Solution via comparing expected utility with search and utility without search 1 pt

TEXT BLOCK

The introduction of the system of poverty alleviation may reduce efficiency, because fewer citizens search for a job. A politician proposes to give the benefit only to citizens that engage in job search. In other words, people only get the benefit if they searched for a job and earn less than 625 euro (either because they did not find a job or because their job pays less than 625 euro). A citizen who did not search for a job, does not receive any benefit (implying that the citizen has an income of 225 euro).

1 PT · OPEN · 1/4 PAGE

Explain (in words) why the politician's proposal to make job search mandatory for receiving benefits could **reduce** efficiency. Use no more than three sentences.

Model answer

If search cost are low, citizens who did not search in the absence of the poverty alleviation system will now be induced to search for a job. However, this is not efficient, as the expected benefits of job search are smaller than the expected gains for these citizens.

Alternative: the cost of monitoring job search are very high.

Beyond the question: note that as $c < 10$, even citizens with $\alpha = 0$ would search for a job:

$$\sqrt{625} - c \geq \sqrt{225} \text{ if } c < 10.$$

correct

1 pt

1 PT · OPEN · 1/5 PAGE

Suppose the politician's proposal to make job search mandatory for receiving benefits **increases** efficiency. From a normative perspective, why could it still be optimal to **not** make job search mandatory? Use no more than two sentences.

Model answer

Mandatory job search imposes cost on low-productivity (or low-income) citizens. Hence, for distributional reasons, it could be optimal to refrain from the job search mandate.

Any other moral-ethical argument is also possible

distributional or moral-ethical argument

1 pt

Efficiency argument

0 pts

Question 5 - Personnel economics #63870249

6 pts · Last updated 10 July, 15:04 · Saved in FEB12001(X) - Toegepaste mi...

TEXT BLOCK

Astrid and Bella are sisters. Each of them has a job in which she can freely decide on her work hours. h_A denotes the number of hours worked by Astrid and h_B denotes the number hours worked by Bella. They earn w per hour. Astrid's utility function is:

$$wh_A + \mu(wh_A - wh_B) - (h_A)^2,$$

where the first term reflects utility from income and the last term reflects the utility costs from loss of leisure. The term in the middle reflects that Astrid finds it important how her income compares to the income of her sister. We assume that $\mu > 0$, and so assume that Astrid likes it when she earns more than her sister and dislikes it to earn less than her. The utility function of Bella is $wh_B + \mu(wh_B - wh_A) - (h_B)^2$. The sisters choose their hours independently from each other and simultaneously.

2 PTS · OPEN · 1/2 PAGE

How many hours will Astrid and Bella choose to work?

Model answer

Write down the utility function and differentiate to h_A and set equal to zero, which gives: $w + \mu w - 2h_A = 0$. Solving gives $h_A = \frac{1}{2} (1+\mu) w$. Likewise for h_B .

Write down the utility func...

2 pts

Write down the utility function and differentiate to h_A and set equal to zero, which gives: $w + \mu w - 2h_A = 0$. Solving gives $h_A = \frac{1}{2} (1+\mu) w$. Likewise for h_B .

2 PTS · OPEN · 1/2 PAGE

Assume now that Astrid and Bella jointly decide on their work hours. They choose work hours so as to maximize the sum of their utilities. How many hours will Astrid and Bella work?

Model answer

The sum of utilities is $wh_A + wh_B - (h_A)^2 - (h_B)^2$. Maximizing to h_A and h_B gives $wh_A - 2h_A = 0$ and $wh_B - 2h_B = 0$. Solving gives $h_A = h_B = \frac{1}{2} w$.

The sum of utilities is whA...

2 pts

The sum of utilities is $wh_A + wh_B - (h_A)^2 - (h_B)^2$. Maximizing to h_A and h_B gives $w - 2h_A = 0$ and $w - 2h_B = 0$. Solving gives $h_A = h_B = \frac{1}{2} w$.

2 PTS · OPEN · 1/2 PAGE

Now assume that Astrid and Bella can not jointly choose their work hours, but can jointly decide to implement an income tax scheme. Any tax receipts will be donated to a charity that Astrid and Bella do not care about. The income tax system can be as simple or complex as they like. Do Astrid and Bella jointly decide to implement a income tax scheme and, if yes, how does it look like? Motivate your answer.

Model answer

Astrid and Bella are best off if they decide to implement a tax rate of 100% for income above the income that corresponds with the number of hours that makes Astrid and Bella most happy, that is $h = \frac{1}{2}w$. The corresponding income is $\frac{1}{2}w * w = \frac{1}{2} w^2$. Income below this level remains untaxed.

Astrid and Bella are best o...

2 pts

Astrid and Bella are best off if they decide to implement a tax rate of 100% for income above the income that corresponds with the number of hours that makes Astrid and Bella most happy, that is $h = \frac{1}{2}w$. The corresponding income is $\frac{1}{2}w * w = \frac{1}{2} w^2$. Income below this level remains untaxed.

Question 6 - Personnel economics #63870282

10 pts · Last updated 10 July, 15:15 · Saved in FEB12001(X) - Toegepaste mi...

TEXT BLOCK

A company employs one worker. The worker has the following utility function:

$$U = bq + \gamma e - \frac{1}{2} \theta e^2$$

where U is the utility of the worker, $b \geq 0$ is the bonus per unit of output, q is the level of output, e the level of effort of the worker, $\gamma > 0$ and $\theta > 0$. The parameter γ can be interpreted as a measure of intrinsic motivation of the worker. The worker aims for maximum utility. The production function is $q = e$. The company sells the output at a price of $p > 0$ per unit of output. We assume that $p > \gamma$. Throughout this exercise, we assume that the worker is better off with this job than with any other job or without a job. The company does not pay a base salary to the worker. The profit of the company is the difference between sales revenues and bonus costs. The company aims for maximum profit. Social welfare is defined as the sum of the utility of the worker and the profit of the company.

2 PTS · OPEN · 1/2 PAGE

Derive the level of effort that maximizes social welfare.

Model answer

Social welfare is equal to $bq + \gamma e - \frac{1}{2} \theta e^2 + pq - bq$. The first and last term cancel. Substituting the production function $q=e$ and maximizing to e gives: $\gamma - \theta e + p = 0$. Solving gives $e = (p + \gamma)/\theta$.

I 1 pt

Social welfare is equal to $bq + \gamma e - \frac{1}{2} \theta e^2 + pq - bq$. The first and last term cancel.

II 1 pt

The first and last term cancel. Substituting the production function $q=e$ and maximizing to e gives: $\gamma - \theta e + p = 0$. Solving gives $e = (p + \gamma)/\theta$.

2 PTS · OPEN · 2/5 PAGE

Derive whether in the above model the following statement is true: If the worker has a higher intrinsic motivation (that is, if γ is higher), then an increase of the bonus has a smaller effect on the level of effort that is chosen by the worker.

Model answer

The worker chooses his effort so as to maximize utility $bq + \gamma e - (\frac{1}{2})\theta e^2$. Substituting the production function $q=e$ and differentiating to e , equating to zero, and solving gives $e = (b + \gamma)/\theta$. The effect of a higher bonus on effort follows from visual inspection (the expression shows that when b goes up by one, effort e goes up by $1/\theta$) or by differentiating the expression for e to b which gives $de/db = 1/\theta$. Hence, the effect is independent of intrinsic motivation γ . The statement is false.

I 1 pt

The worker chooses his effort so as to maximize utility $bq + \gamma e - (\frac{1}{2})\theta e^2$. Substituting the production function $q=e$ and differentiating to e , equating to zero, and solving gives $e = (b + \gamma)/\theta$.

II 1 pt

The effect of a higher bonus on effort follows from visual inspection (the expression shows that when b goes up by one, effort e goes up by $1/\theta$) or by differentiating the expression for e to b which gives $de/db = 1/\theta$. Hence, the effect is independent of intrinsic motivation γ . The statement is false.

4 PTS · OPEN · 4/5 PAGE

Derive what level of the bonus the company chooses. Note: You can get four points for this!

Model answer

The company chooses the bonus b so as to maximize profit $(pq - bq)$. We know from the production function that $q = e$. The worker chooses effort so as to maximize utility $bq + \gamma e - (\frac{1}{2})\theta e^2$. Substituting the production function $q=e$ and differentiating to e , equating to zero, and solving gives $e = (b + \gamma)/\theta$ and so $q = (b + \gamma)/\theta$. Substituting into the profit function and differentiating to b and equating to zero gives: $p/\theta - 2b/\theta - \gamma/\theta = 0$. Solving yields: $b = (\frac{1}{2})(p - \gamma)$.

I 1 pt

The company chooses the bonus b so as to maximize profit $(pq - bq)$.

II 1 pt

We know from the production function that $q = e$. The worker chooses effort so as to maximize utility $bq + \gamma e - (\frac{1}{2})\theta e^2$. Substituting the production function $q=e$ and differentiating to e , equating to zero, and solving gives $e = (b + \gamma)/\theta$ and so $q = (b + \gamma)/\theta$.

III 1 pt

Substituting into the profit function and differentiating to b and equating to zero gives: $p/\theta - 2b/\theta - \gamma/\theta = 0$.

IV 1 pt

Solving yields: $b = (\frac{1}{2})(p - \gamma)$.

2 PTS · OPEN · 3/10 PAGE

Explain in words why the profit-maximizing level of the bonus depends on the level of intrinsic motivation γ . Note: we ask for economic reasoning here, not for a mathematical derivation or mathematical reasoning. Use at most three sentences.

Model answer

If the worker is more strongly motivated, an increase in the bonus is more costly for the company, because it also needs to pay the bonus increase for the production that is driven by intrinsic motivation. Intrinsic motivation therefore makes the profit-maximizing level of the bonus lower. This can be seen from the third term of the first-order condition for maximum profit.

I 1 pt

If the worker is more strongly motivated, an increase in the bonus is more costly for the company, because it also needs to pay the bonus increase for the production that is driven by intrinsic motivation.

II 1 pt

Intrinsic motivation therefore makes the profit-maximizing level of the bonus lower. This can be seen from the third term of the first-order condition for maximum profit.

Question 7 - Personnel economics #63870286

4 pts · Last updated 26 June, 08:49 · Saved in FEB12001(X) - Toegepaste mi...

TEXT BLOCK

A municipality is searching for workers to produce public goods. In total, the municipality would like to reach a production of size Q . There are two types of workers in the labor market: motivated and unmotivated workers. Motivated workers have the following utility function:

$$U = w + \gamma e - \theta e^2$$

where U is utility, w the base salary, e the level of effort, and $\gamma > 0$ a positive parameter.

Unmotivated workers have the following utility function:

$$U = w - \theta e^2$$

We assume that the municipality cannot pay any bonuses. If a worker decides not to work at the municipality, then he or she achieves utility level U^{alt} . This holds for both the motivated and unmotivated workers. The production level achieved by a worker is equal to his or her effort level. Finally, we assume that if somebody applies at the municipality, the municipality can observe whether it is a motivated or unmotivated worker. There is a large number of both types of workers present in the labor market, and so if the municipality wants to use workers of one single type, then there are sufficiently many available to produce production Q .

1 PT · OPEN · 3/10 PAGE

Calculate how much production an unmotivated worker produces if he is hired by the municipality.

Model answer

An unmotivated worker maximizes utility $w - \theta e^2$ to e , which results in first-order condition: $-2\theta e = 0$. Hence, an unmotivated worker chooses $e = 0$ and because $q = e$ this also means that $q = 0$.

Fully correct

1 pt

Fully correct

1 PT · OPEN · 3/5 PAGE

Show that if the parameter γ is higher, that then a motivated worker chooses a higher level of effort and is willing to work for a lower base salary.

Model answer

A motivated worker maximizes utility $w + \gamma e - \theta e^2$ to e , resulting in the first-order condition: $\gamma - 2\theta e = 0$. Hence, effort is equal to $e = \gamma / 2\theta$. The base salary at which a motivated worker is willing to take the job follows from the participation constraint, which reads: $w + \gamma e - \theta e^2 \geq U^{alt}$. Substituting $e = \gamma / 2\theta$ and rewriting gives $w \geq U^{alt} - \gamma^2 / 4\theta$. The minimum salary thus declines in γ .

Fully correct

1 pt

Fully correct

1 PT · OPEN · 7/20 PAGE

How many motivated workers are needed to achieve production level Q ?

Model answer

Each motivated worker chooses $e = \gamma/2\theta$ and so contributes $\gamma/2\theta$ to total output. To produce Q , one needs $Q/(\gamma/2\theta)$ motivated workers. Rewriting gives $2\theta Q/\gamma$ (but rewriting is not necessary for the full point).

Fully correct

1 pt

Fully correct

TEXT BLOCK

Assume now that the municipality cannot distinguish between the two types of workers during the application procedure.

1 PT · OPEN · 3/5 PAGE

Does this lack of information cause a problem for the municipality? If not, explain why not. If yes, then explain what problem arises.

Model answer

No, because setting the base salary at minimum $U^{alt} = \gamma^2/4\theta$ and below U^{alt} , makes that the motivated workers are willing to take the job and the unmotivated do not. It is therefore no problem that the municipality cannot distinguish types. By setting the wage at the right level, it can induce self-selection of the motivated workers.

Fully correct

1 pt

Fully correct

Question 8 - Personnel economics #63870297

6 pts · Last updated 26 June, 08:52 · Saved in FEB12001(X) - Toegepaste mi...

TEXT BLOCK

The HR manager of a large retail company consisting of 900 stores wonders whether the personnel policies can be improved. Currently, store managers have a lot of autonomy in choosing the personnel policies in their store. The HR manager collected data and found out that 400 store managers currently use performance pay for workers whereas the other 500 store managers do not.

2 PTS · OPEN · 3/10 PAGE

A personnel economist argues that, according to theory, stores that use performance pay for workers may achieve higher sales and profits for at least two reasons, one related to moral hazard and another related to adverse selection. Explain to the HR manager in two clear sentences what the personnel economist tries to tell.

Performance pay can encoura...

1 pt

Performance pay can encourage workers to exert more effort because more effort may result in higher performance and hence higher pay for workers.

Performance pay may help to...

1 pt

Performance pay may help to attract more productive workers because they expect to perform better and hence earn more.

2 PTS · OPEN · 3/10 PAGE

A data scientist had a look at the sales and profits data and found out that the 400 stores that use performance pay for workers do significantly worse than the 500 stores that do not. Give one possible reason for this pattern in the data. Use at most two sentences. You can use technical terms, but you need to explain them.

It could be that store mana...

2 pts

It could be that store managers only revert to using performance pay when their store is not performing well. If the store is performing really well already and is close to its capacity constraint (lots of customers who are all very much willing to buy already), performance pay may not bring much. Another possible reason is that performance pay crowds out intrinsic motivation. If this effect is strong, it dominates the standard incentive effect and performance may go down.

2 PTS · OPEN · 1/2 PAGE

The personnel economist proposes to run a field experiment to find out whether sales and profits would increase if performance pay for workers was abandoned in the 400 stores that use them. Sketch in at most three sentences how such a field experiment should be designed so as to provide credible evidence of the effect of performance pay for workers on sales and profits.

Randomly assign half of the...

1 pt

Randomly assign half of the 400 stores to a treatment group and half to a control group.

Abandon performance pay for...

1 pt

Abandon performance pay for workers in the stores in the treatment store and keep it in the control stores. Compare sales and profits over time between treatment group and control group.

Question 9 - Personnel economics #63870298

4 pts · Last updated 24 June, 12:57 · Saved in FEB12001(X) - Toegepaste mi...

TEXT BLOCK

A not-for-profit organization aims to maximize output. To produce output, the organization hires workers. Each worker contributes q to total output. If the organization offers wage W , then the organization hires $N(W)$ workers, where $N'(W) > 0$. The not-for-profit has a budget equal to B .

2 PTS · OPEN · 9/20 PAGE

Derive by how much the total expenditures of the organization increase when it offers a one euro higher wage.

Total cost of the organizat... 1 pt

Total cost of the organization are $WN(W)$.

Taking the derivative to W ... 1 pt

Taking the derivative to W gives the effect of a one euro increase of the wage on total cost: $N(W) + WN'(W)$.

2 PTS · OPEN · 7/20 PAGE

Under what condition does this organization classify as an organization with monopsony power?

When $N'(W)$ is smaller than ... 2 pts

When $N'(W)$ is smaller than plus infinity. Under competition, a slight decrease in the wage results in all workers wanting to quit. Whenever $N'(W)$ is smaller than infinity, it holds that the organization can reduce the wage without losing all of its workers, i.e. it has (some) monopsony power.

Additional answer space #50629722

0 pts · Last updated 24 June, 12:38

TEXT BLOCK

Only use this extra space if you made a mistake in one of your answers. Please do not forget to **indicate in the answer box of that particular question where the correct answer can be found.**

Otherwise, the answers in the additional space below will not be reviewed.

NB: Do **not** use this additional space for answers that are too long.

OPEN · 4/5 PAGE

Additional answer box 1

OPEN · 1/2 PAGE

Additional answer box 2

OPEN · 1/2 PAGE

Additional answer box 3

OPEN · 1/2 PAGE

Additional answer box 4

OPEN · 1/2 PAGE

Additional answer box 5